A-A-53864
16 March 1989

COMMERCIAL ITEM DESCRIPTION

REGULATOR, NITROGEN, SINGLE PORT

The General Services Administration has authorized the use of this commercial item description.

This commercial item description covers a single port nitrogen regulator suitable for use with air-powered surgical instruments.

Salient characteristics:

General. The regulator shall be a single port nitrogen regulator in accordance with applicable portions of Compressed Gas Association (CGA) Pamphlet E-7. The regulator shall be capable of operating on compressed air provided it has been passed through a filter whose openings are no more than 66 microns in size and dried to -40°F dew point.

The regulator shall have a twist knob to control outlet operating pressure, a nitrogen cylinder pressure gauge, and an outlet operating pressure gauge. The regulator shall have a female Schrader fitting for connection to a hose suitable for delivering the nitrogen to the instrument being operated. The regulator shall have a sintered metal inlet gas filter.

The cylinder pressure gauge shall conform to applicable portions of Underwriters Laboratories' (UL) Standards 404. Proof of such conformance shall consist of either a UL label or trademark affixed to the gauge or a laboratory report from a nationally recognized independent laboratory. If an independent laboratory is used, it must be adequately equipped and competent to determine compliance with UL standards. The Government reserves the right to require a facilities report and to conduct a facilities survey.

Material. The regulator body, inlet port nipple and hexagon nut, and safety pressure relief and outlet port adapters shall be fabricated from chromium plated over nickel plated brass. Internal surfaces of the regulator body should not be plated. The outlet pressure adjusting knob shall be fabricated of rigid plastic.

AMSC N/A FSC 6515

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Both gauge bodies shall be fabricated from chromium plated over nickel plated steel and both face covers shall be fabricated from clear plastic. The inlet filter shall be fabricated of sintered bronze.

Style, design, and dimensions. The regulator shall have a gas inlet connection fitting in accordance with Compressed Gas Association (CGA) Standard V.1 or American National Standards Institute (ANSI) B57.1, Connection No. 580. The operating pressure gauge shall read 0 to 300 psi and the cylinder storage pressure gauge shall read 0 to 4000 psi. The outlet port shall have a Schrader-type female connector. The safety relief valve shall be set to release at 200 psi. The inlet filter shall be assembled with filter openings no greater than 66 microns in size. The regulator shall be in accordance with Figure 1.

Performance. When tested as specified herein, the operating pressure gauge shall return to the no flow position when the pressure is released from the regulator. The regulator shall exhibit no leakage as evidenced by bubble formation when tested as specified herein. With the nitrogen cylinder pressure gauge at 4000 psi, the regulator shall be able to fully operate an air-powered instrument at 100 psi.

Identification marking. Both pressure gauges and/or the regulator shall be clearly and legibly marked with the manufacturer's name or registered trademark.

Workmanship. The nitrogen regulator shall be free from defects which detract from its appearance or impair its serviceability.

Unit. Each (EA). One nitrogen regulator, as specified, constitutes one unit.

Quality Assurance Provisions.

Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

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Records. Records of examinations and tests performed by or for the contractor shall be maintained by the contractor and made available to the Government upon the Government's request, at any time, or from time to time, during the performance of the contract and for a period of three years after delivery of the supplies to which such records relate.

Inspection. Inspection, as used herein, is defined as both examination (such as visual or auditory investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties) of the item.

Examination. Nitrogen regulators shall be examined to determine compliance with all requirements contained in this specification.

Sampling for examination. Examination shall be conducted in accordance with MIL-STD-105 with an AQL of 1.5% and an inspection level of II.

Tests. Tests shall be conducted to determine compliance with specification requirements. Where feasible, the same sample shall be used for the determination of two or more test characteristics.

Operating pressure gauge test. On a nitrogen cylinder containing at least 2000 psi, install a precision reducing device capable of regulating nitrogen gas pressure. Outlet from this device is the input to the regulator being tested. Install regulator being tested. Set pressure to the regulator being tested at 2000 psi and set operating pressure at 100 psi. Release gas pressure in the regulator being tested. Repeat at 1000 and 500 psi.

Leakage test. Install the regulator being tested on a nitrogen cylinder capable of delivering at least 2000 psi. With 2000 psi being applied to the regulator, turn out the operating pressure knob fully and apply a film of grease-free soapy water to all connections. Next turn the operating pressure knob in all the way or until the operating pressure is 190 psi, (whichever occurs first). Check connections with soapy water, especially the female Schrader fitting.

Sampling for tests. Sampling for tests shall be conducted in accordance with MIL-STD-105 and specified herein. Sampling shall be level S-1, AQL 1.0%. The unit of product for test purposes shall be one nitrogen regulator.

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Contractor certification. The contractor shall certify that the product offered meets the salient characteristics of this description and conforms to the producers' own drawings, specifications, standards, and quality assurance practices. The Government reserves the right to require proof of such conformance prior to first delivery and thereafter as may be otherwise provided for under the provisions of the contract.

Regulatory requirements.

Federal Food, Drug and Cosmetic Act. If the product covered by this document has been determined by the U.S. Food and Drug Administration to be under its jurisdiction, the offeror/contractor shall comply, and be responsible for compliance by its subcontractors/suppliers, with the requirements of the Federal Food, Drug and Cosmetic Act, as amended, and regulations promulgated thereunder. In addition, the offeror/contractor shall comply, and be responsible for compliance by its subcontractors/suppliers, with the requirements of all other applicable Federal, State, and local statutes, ordinances, and regulations.

Recovered materials. The offeror/contractor is encouraged to use recovered material in accordance with Public Law 94-580, as amended, and as implemented by Federal Acquisition Regulation Subpart 23.4 to the maximum extent practical.

Preservation, packaging, packing, labeling, and marking. Unless otherwise specified, preservation, packaging, and packing shall be to a degree of protection to preclude damage to containers and/or contents thereof under normal shipping conditions, handling, etc., involving shipment from the supply source to the receiving activity, plus reshipment from receiving activity, and shall conform to applicable carrier's rules and regulations. Intermediate and exterior package quantities and labeling and marking shall be as specified in the contract and/or order.

NOTE: The following National Stock Number is covered by this document:

NSN	ITEM IDENTIFICATION
6515-01-267-6114	Regulator, Nitrogen, with Schrader Female Connector, CGA Fitting

Ordering data (Intermediate/exterior package quantities, labeling, and marking must be specified in the contract and/or order).

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MILITARY INTERESTS:

Custodians:

Army - MD
Navy - MS
Air Force - 03

PREPARING ACTIVITY:

DoD - MB

Agent:

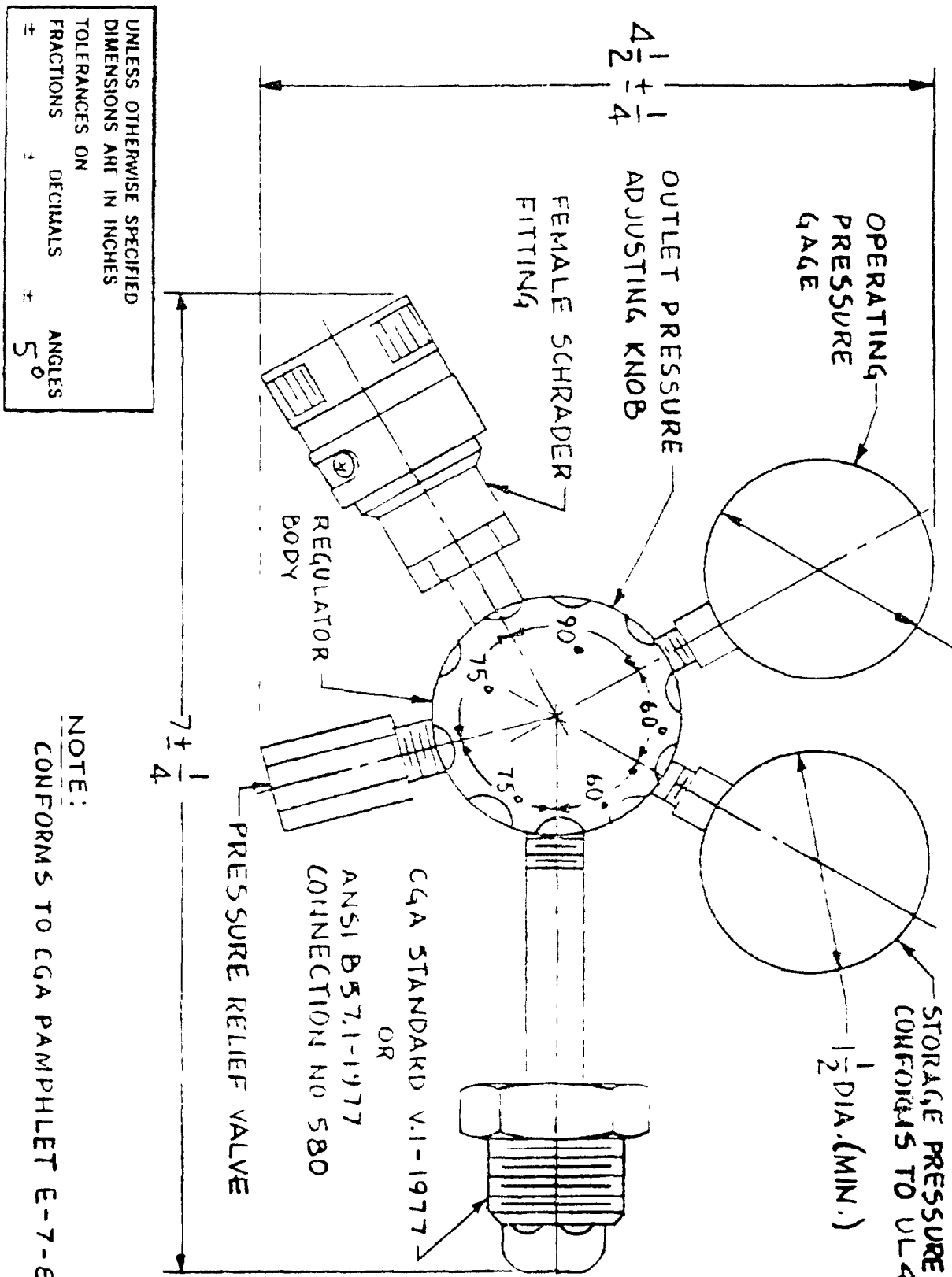
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CIVIL AGENCY COORDINATING ACTIVITIES:

VA-OSS
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FDA-MPQAS

Project No. 6515-4456

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NOTE:
CONFORMS TO CGA PAMPHLET E-7-83.

FIGURE 1 REGULATOR, NITROGEN, SINGLE PORT

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1 DOCUMENT NUMBER	2 DOCUMENT TITLE
A-A-53864	REGULATOR, NITROGEN, SINGLE PORT

3a NAME OF SUBMITTING ORGANIZATION	4 TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
	☐ USER
b ADDRESS *(Street, City, State, ZIP Code)*	☐ MANUFACTURER
	☐ OTHER *(Specify)*

5 PROBLEM AREAS

a. Paragraph Number and Wording

b. Recommended Wording

c. Reason/Rationale for Recommendation

6 REMARKS

7a NAME OF SUBMITTER *(Last, First, MI)* – Optional	b WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8 DATE OF SUBMISSION *(YYMMDD)*

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DD FORM 1426

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NOTE. This form may not be used to request copies of documents, nor to request waivers, deviations, or clarification of specification requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

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